

Spotify 2023 Data Analysis – Business Intelligence Project

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This report documents the implementation of the supplied Spotify 2023 Power BI Project (PBIP). The project converts the Spotify song dataset into an interactive six-page dashboard covering platform comparison, artist impact, playlist and chart performance, Shazam discovery, Major versus Minor characteristics, and BPM performance.

1. Project Objective

- Import and clean the Spotify 2023 dataset.
- Prepare the data for reliable numeric aggregation.
- Create reusable DAX measures and BPM calculated columns.
- Build interactive filters and six analytical report pages.
- Compare streams with playlists, charts, artists, Shazam activity, musical mode and BPM.

2. Step-by-Step Implementation – Part 1

1. Extract the supplied ZIP file into a local folder.
2. Open Microsoft Power BI Desktop.
3. Open the file Spotify_2023_Data_Analysis_MST.pbip. A PBIP project stores report and semantic-model definitions in project folders; the .pbip file points Power BI Desktop to the report. [↗cite↗turn0search0↗](#)
4. If PBIP support is not enabled in the installed Power BI version, open File → Options and settings → Options → Preview features and enable the Power BI Project (.pbip) save option. [↗cite↗turn0search0↗](#)
5. Open the report and verify that the source table is named spotify-2023.

Project source

The supplied CSV contains 953 song records and 24 columns. There are 952 valid stream values and 1 invalid stream value handled during cleaning.

3. Main Project Components

Component	Purpose
spotify-2023.csv	Original Spotify song dataset.
Spotify_2023_Data_Analysis_MST.pbip	Power BI Project entry point.
Semantic Model	Stores fields, measures and calculated columns.
Report definition	Stores the six dashboard pages and visuals.

Implementation Steps – Data Preparation, Model and Dashboard

4. Step-by-Step Implementation – Part 2

6. In Power BI Desktop, select Home → Transform data to open Power Query.
7. Check that the CSV data has been loaded into the spotify-2023 table.
8. Remove comma characters from numeric platform-count fields such as Spotify playlists, Spotify charts, Apple playlists, Apple charts, Deezer playlists, Deezer charts and Shazam charts.
9. Set the correct data types: text fields as Text, counts/years/BPM as Whole Number, and audio-feature percentages as numeric values.
10. Convert the streams column to a numeric whole-number type.
11. Replace the streams conversion error with null. Do not replace it with zero because an invalid value is not evidence of zero streams.
12. Select Close & Apply to load the cleaned data into the Power BI model.
13. Create the reusable DAX measures for Total Streams, Average Streams, Total Songs, Average Danceability, Average Energy, Average BPM, playlist exposure, chart presence, Average Valence and Average Acousticness.
14. Create the BPM Range calculated column with six groups: <90, 90-109, 110-129, 130-149, 150-169 and 170+.
15. Create BPM Range Order with values 1–6 and use it to sort BPM Range in the correct order.
16. Add slicers for released_year, artist(s)_name, mode and BPM Range where required.
17. Create the six report pages and add the required charts, cards, scatter plots and detail tables.
18. Add page-navigation buttons so the user can move between all six analytical pages.
19. Apply Top N = 10 filtering to the artist charts on the Artist Impact page.
20. Refresh the report and test every slicer, chart and navigation button.
21. Save the completed report as PBIX if a single packaged Power BI file is required, and export the finished report to PDF.

5. Important DAX Measures

Measure	Purpose
Total Streams	SUM of streams
Average Streams	Average stream count
Total Songs	Number of song records
Spotify Playlist Exposure	SUM of Spotify playlist counts
Apple Playlist Exposure	SUM of Apple playlist counts
Deezer Playlist Exposure	SUM of Deezer playlist counts
Spotify Chart Presence	SUM of Spotify chart counts
Apple / Deezer Chart Presence	SUM of corresponding chart counts
Shazam Chart Presence	SUM of Shazam chart counts
Average Danceability / Energy	Average audio characteristics
Average BPM	Average tempo
Average Valence / Acousticness	Average musical characteristics

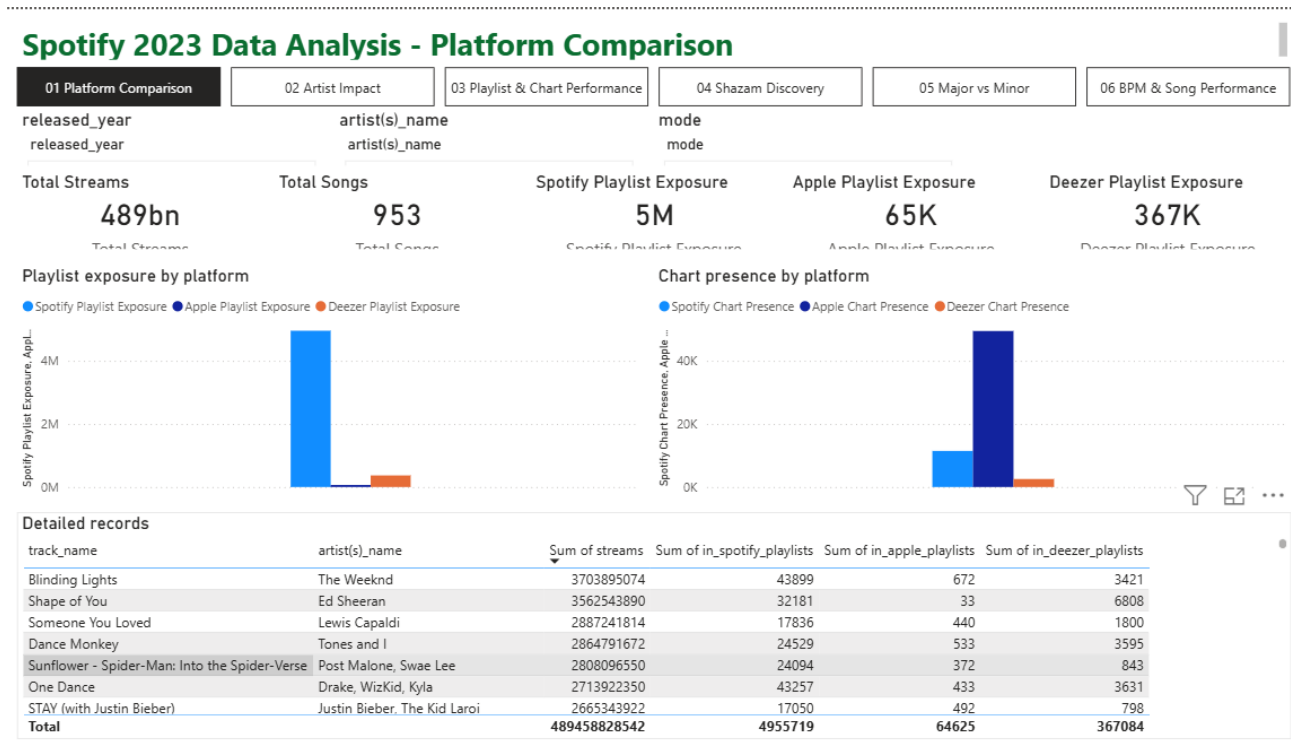
Model result

The semantic model in the supplied project contains 15 measures and two BPM-related calculated columns. The dashboard visuals reuse these model calculations so filtering remains consistent.

6. Power BI Project Format

The PBIP format separates report and semantic-model definitions into project folders and is designed for development and source-control workflows. Microsoft documents the .pbip file as a pointer to the report folder. [?cite?turn0search0?](#)

Dashboard Explanation – Platform Comparison



Screenshot 1 – Spotify 2023 Data Analysis – Platform Comparison

What this screenshot shows

Screenshot 1 is the main platform-comparison dashboard. It provides a high-level view of total streams, total songs, playlist exposure and chart presence across Spotify, Apple and Deezer.

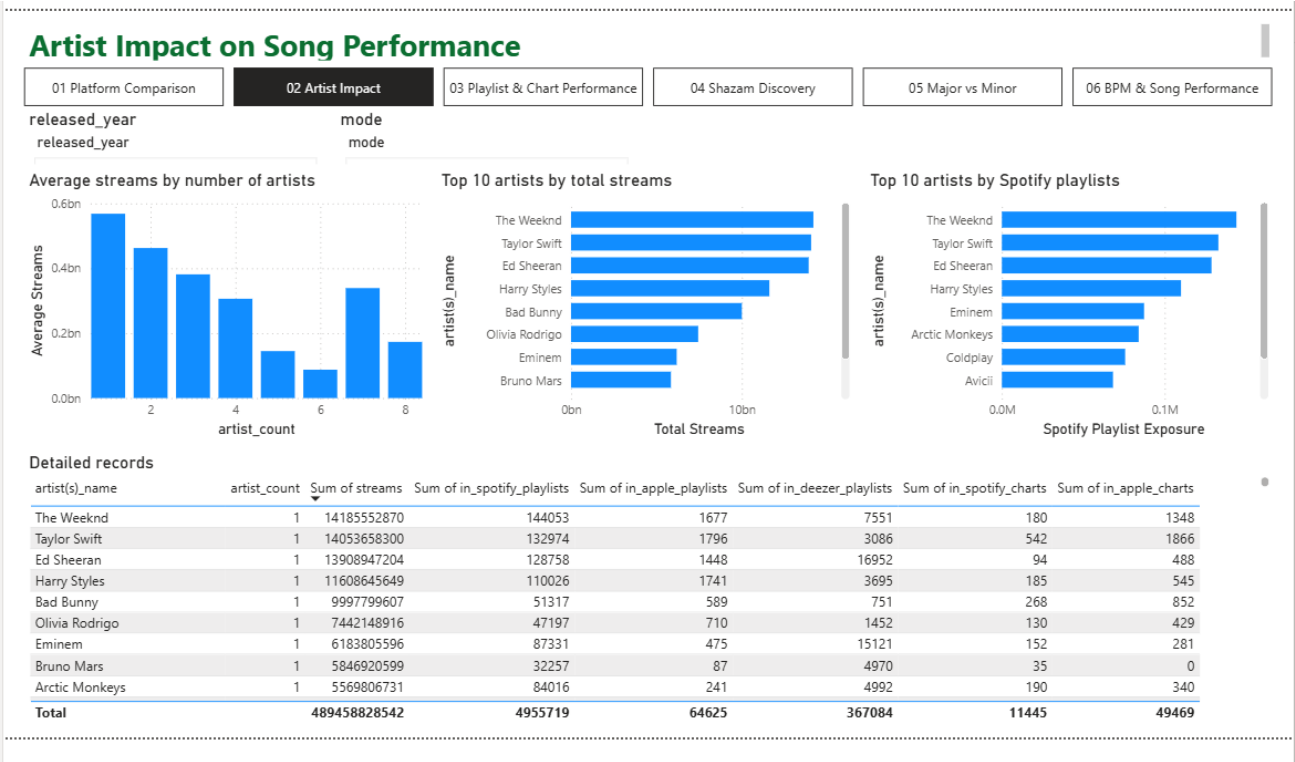
How to read the screen

- The top navigation buttons move between the six report sections.
- The released_year and artist(s)_name filters allow the analysis to be narrowed to a selected year or artist.
- The KPI cards summarize Total Streams, Total Songs and playlist exposure for the selected filter context.
- The left chart compares playlist exposure by platform.
- The right chart compares chart presence by platform.
- The Detailed records table provides the song-level records behind the summary visuals.

Interpretation

- The report contains 953 song records and about 489.5 billion numeric streams.
- Spotify playlist exposure is visibly much larger than Apple and Deezer playlist exposure in the displayed dataset.
- The platform metrics are useful for comparison, but the platforms may use different counting conventions.
- The detailed table allows the user to move from overall platform totals to individual songs and artists.

Dashboard Explanation – Artist Impact



Screenshot 2 – Artist Impact on Song Performance

What this screenshot shows

Screenshot 2 studies artist involvement and artist-level performance. It combines average streams by artist count with Top 10 artist charts and a detailed record table.

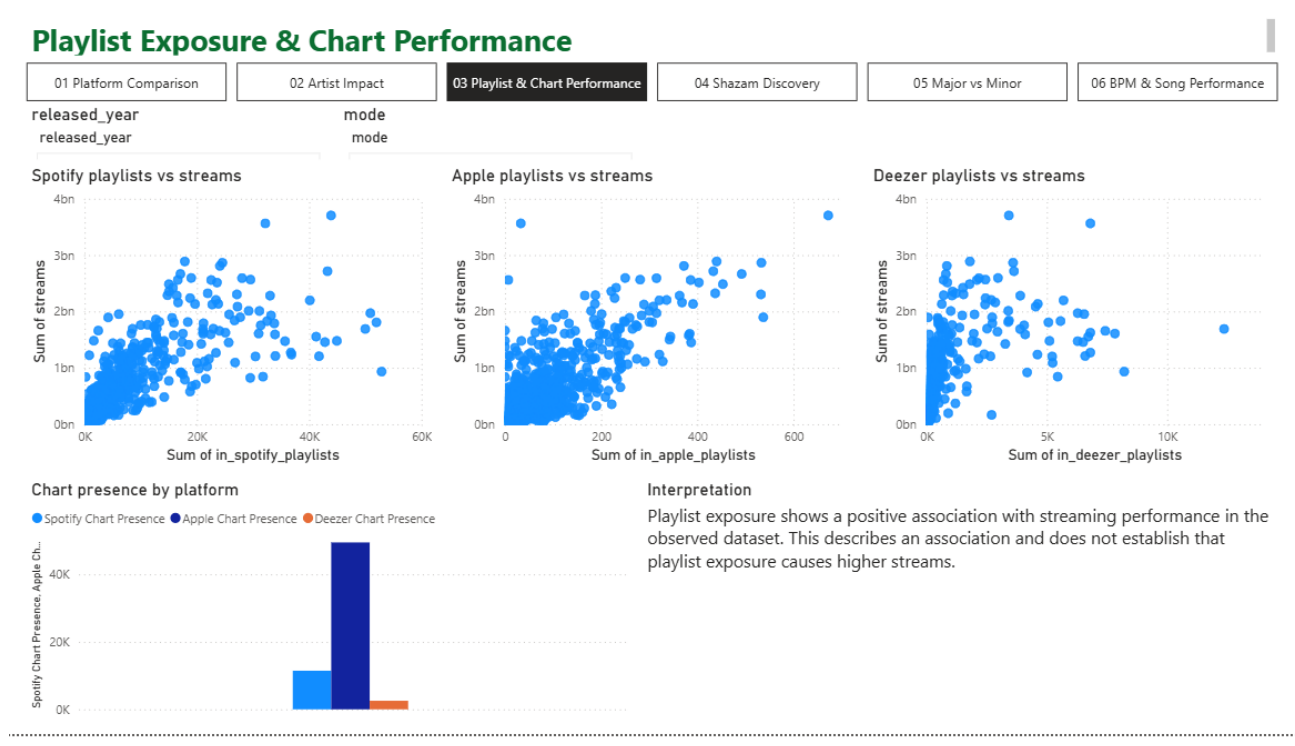
How to read the screen

- Average streams by number of artists compares the mean streaming result for different artist-count groups.
- Top 10 artists by total streams identifies the artists with the largest sum of streams in the dataset.
- Top 10 artists by Spotify playlists identifies artists with the highest accumulated Spotify playlist exposure.
- The detailed table provides streams and playlist/chart measures for individual artists.
- The mode slicer can be used to compare Major and Minor records while the page remains interactive.

Interpretation

- The displayed Top 10 stream ranking begins with The Weeknd, followed by Taylor Swift and Ed Sheeran.
- The highest-performing artists by streams and playlist exposure overlap substantially, although the two metrics are not identical.
- Artist count is an observed grouping variable; differences between groups should not be interpreted as proof that adding artists causes higher or lower streams.
- The table is useful for validating the values behind the Top 10 charts.

Dashboard Explanation – Playlist Exposure & Chart Performance



Screenshot 3 – Playlist Exposure & Chart Performance

What this screenshot shows

Screenshot 3 compares playlist exposure with streaming performance separately for Spotify, Apple and Deezer. Scatter plots are used because each point represents an individual observation and the distribution can be examined directly.

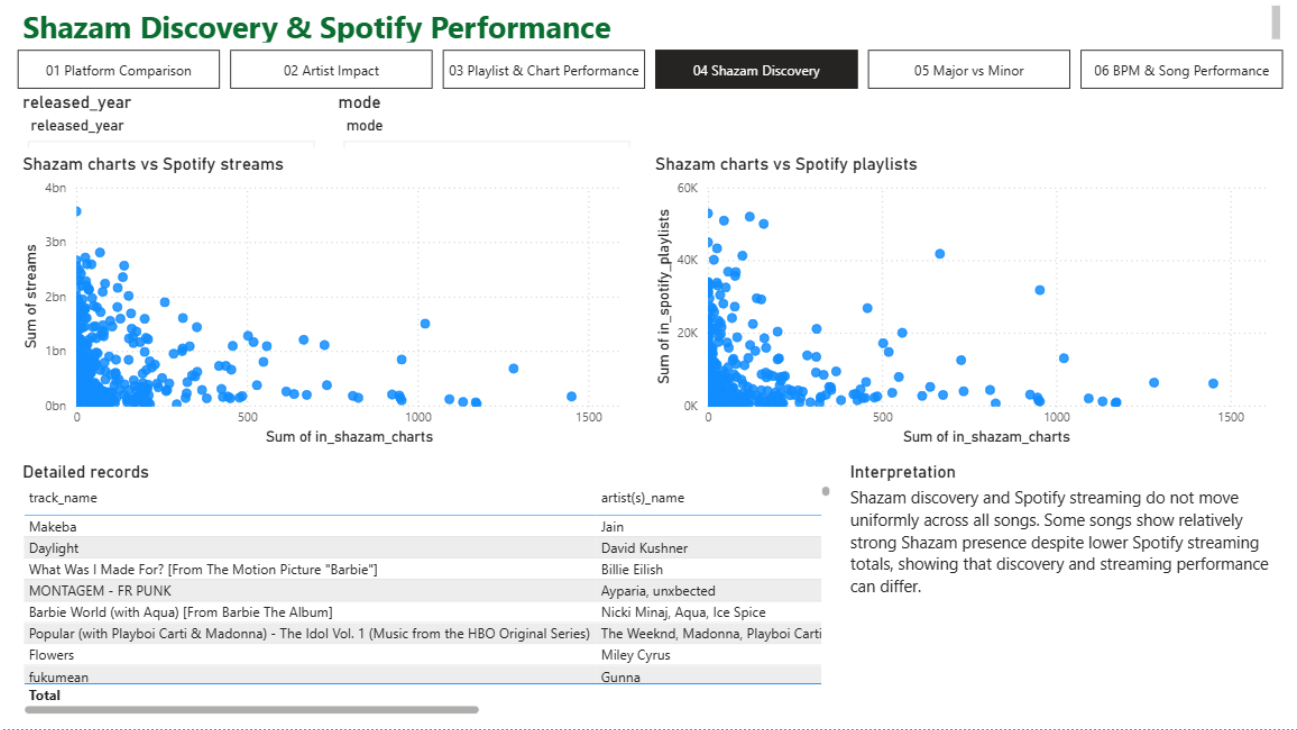
How to read the screen

- Spotify playlists vs streams places Spotify playlist exposure on the horizontal axis and streams on the vertical axis.
- Apple playlists vs streams repeats the analysis for Apple playlist exposure.
- Deezer playlists vs streams repeats the analysis for Deezer playlist exposure.
- The chart-presence visual compares accumulated chart counts across the platforms.
- The interpretation box correctly describes the relationship as an association rather than a causal effect.

Interpretation

- Correlation with streams is approximately 0.79 for Spotify playlists, 0.77 for Apple playlists and 0.60 for Deezer playlists.
- The scatter plots show that songs with greater playlist exposure often occur with higher stream totals, but the points are widely dispersed.
- A relationship in the observed data does not prove that playlist placement itself caused the stream total.
- The three plots make it possible to compare how the same concept behaves across different platforms.

Dashboard Explanation – Shazam Discovery



Screenshot 4 – Shazam Discovery & Spotify Performance

What this screenshot shows

Screenshot 4 examines Shazam chart activity as a discovery-related signal and compares it with Spotify streams and Spotify playlist exposure.

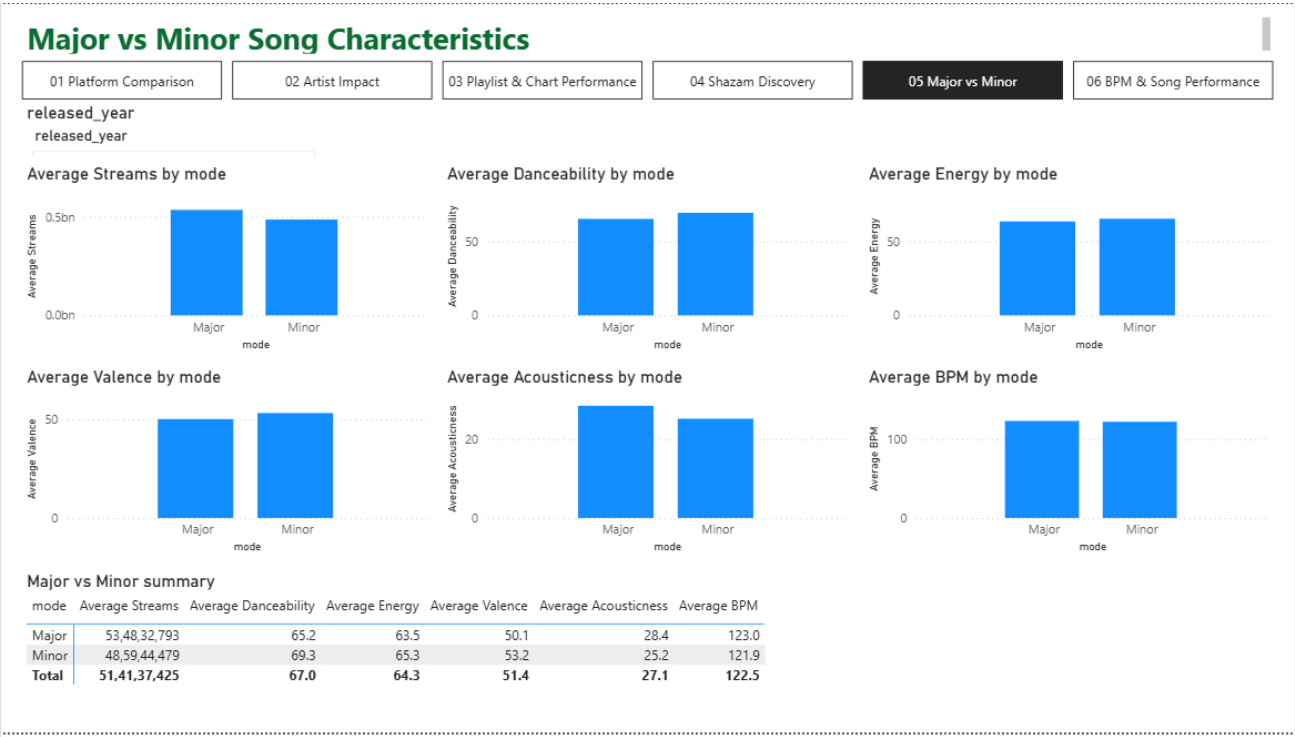
How to read the screen

- The first scatter plot compares Shazam chart presence with Spotify streams.
- The second scatter plot compares Shazam chart presence with Spotify playlist exposure.
- The horizontal axis in each scatter plot represents Shazam chart activity.
- The vertical axis represents the corresponding Spotify performance measure.
- The Detailed records table lists song and artist information for closer inspection.
- The interpretation text emphasizes that discovery and streaming performance can differ.

Interpretation

- The cleaned dataset gives a correlation of approximately 0.02 between Shazam chart presence and streams.
- The scatter distribution is highly dispersed, so Shazam activity does not uniformly correspond to a particular Spotify streaming level.
- A song may receive discovery attention without having the highest Spotify stream total.
- This page is useful for comparing discovery behavior with consumption behavior.

Dashboard Explanation – Major vs Minor



Screenshot 5 – Major vs Minor Song Characteristics

What this screenshot shows

Screenshot 5 compares songs classified as Major or Minor across streams and five musical characteristics: danceability, energy, valence, acousticness and BPM.

How to read the screen

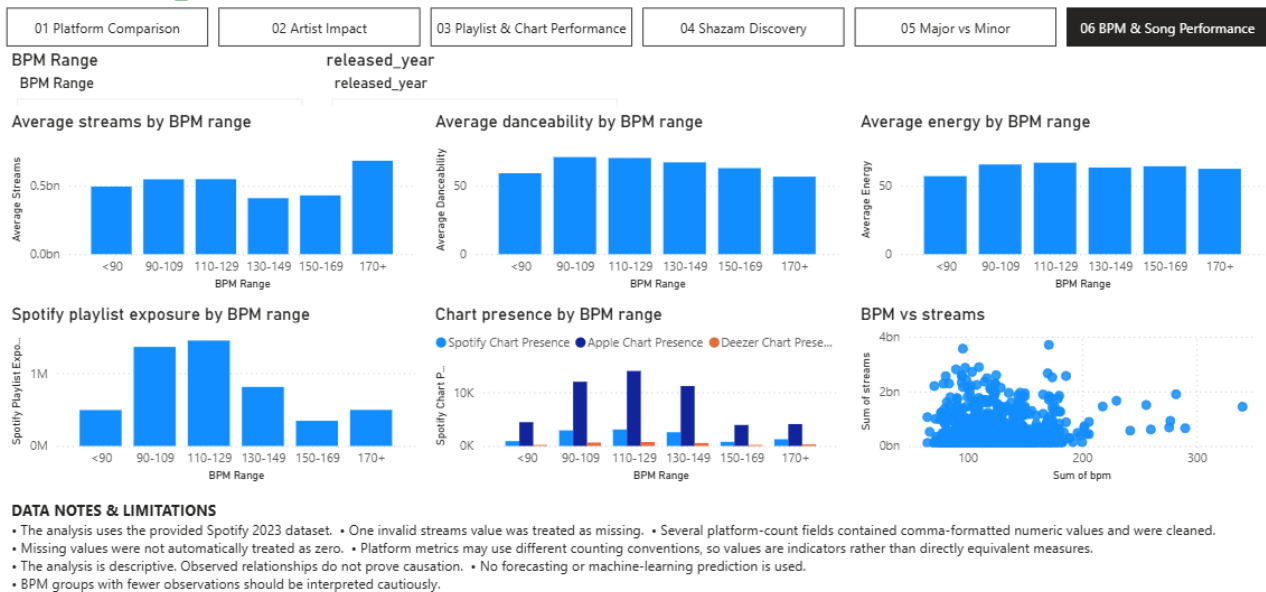
- Average Streams by mode compares mean streams for Major and Minor songs.
- Average Danceability by mode compares the mean danceability score.
- Average Energy by mode compares the mean energy score.
- Average Valence by mode compares the mean valence score.
- Average Acousticness by mode compares the mean acousticness score.
- Average BPM by mode compares average tempo.
- The summary table at the bottom places all six measures side by side.

Interpretation

- Major songs average about 534.8 million streams, while Minor songs average about 485.9 million.
- Minor songs have higher average danceability (69.3) than Major songs (65.2).
- Minor songs have slightly higher average energy (65.3) than Major songs (63.5).
- Major songs have higher average acousticness (28.4) than Minor songs (25.2).
- Average BPM values are close, so the mode groups do not show a large difference in average tempo.

Dashboard Explanation – BPM & Song Performance

BPM & Song Performance



Screenshot 6 – BPM & Song Performance

What this screenshot shows

Screenshot 6 groups songs into six BPM ranges and compares average streams, danceability and energy, plus Spotify playlist exposure and chart presence. The final scatter plot examines BPM against streams.

How to read the screen

- The six BPM groups are <90, 90-109, 110-129, 130-149, 150-169 and 170+.
- Average streams shows the mean stream total within each BPM group.
- Average danceability and energy show how musical characteristics change across tempo groups.
- Spotify playlist exposure and chart presence compare platform visibility by BPM range.
- BPM vs streams plots individual observations rather than only group averages.
- The Data Notes & Limitations section records the cleaning decisions and cautions against causal interpretation.

Interpretation

- The highest average-stream BPM group in the cleaned dataset is 170+, at about 684.4 million average streams.
- The highest Spotify playlist exposure occurs in the 110-129 BPM group.
- The highest Spotify chart presence occurs in the 110-129 BPM group.
- The 170+ BPM group should be interpreted cautiously because it contains fewer observations than several central BPM groups.
- The scatter plot shows that BPM alone does not determine streaming performance; many other song and platform factors are involved.

Final Project Conclusion

The completed Power BI project demonstrates the full Business Intelligence workflow: source-data preparation, cleaning, data-type correction, semantic modeling, DAX calculations, interactive filtering, visualization and report interpretation. The six screenshots document the final dashboard output, while the implementation steps explain how the report is constructed.